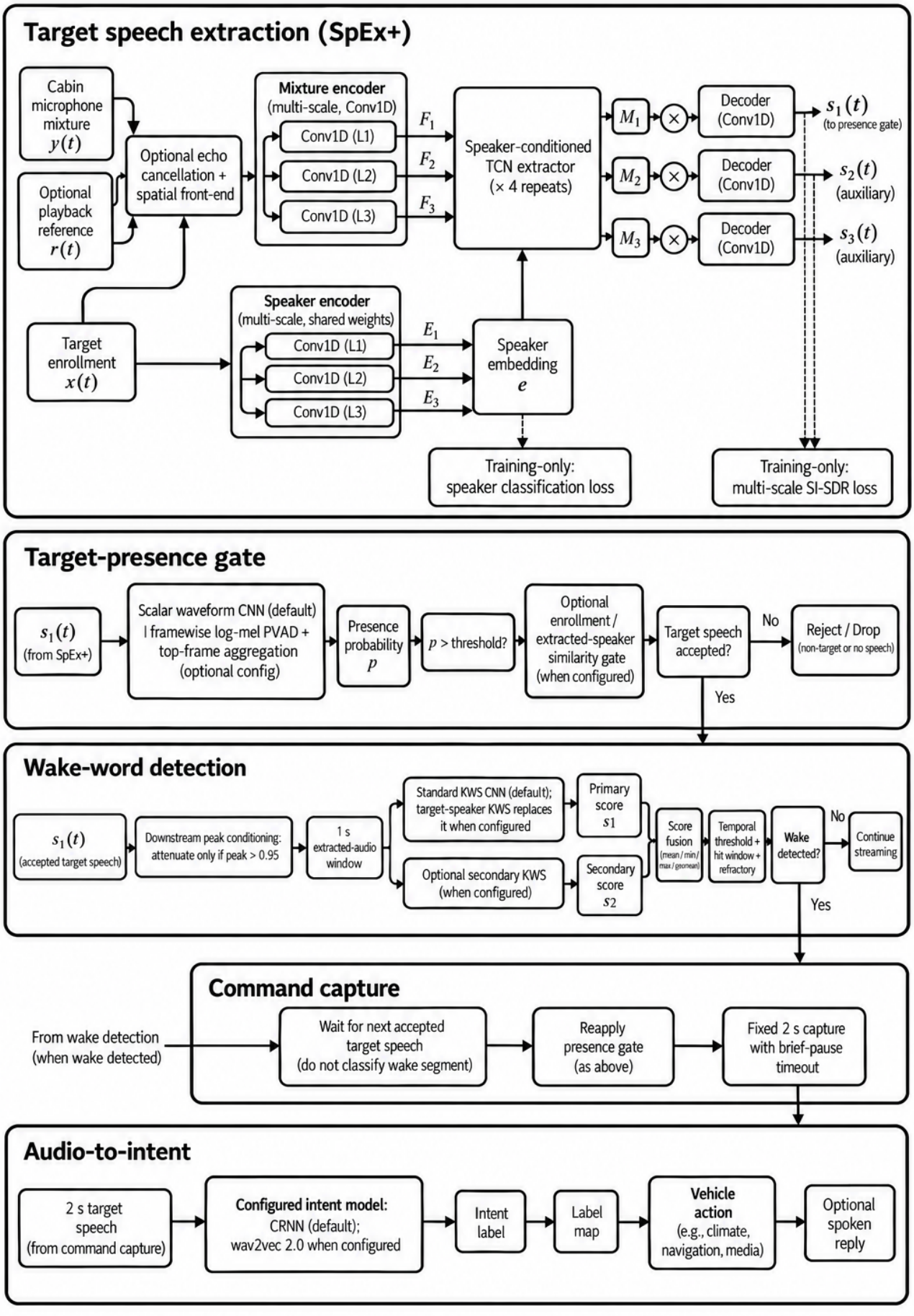

Target speech extraction (SpEx+)
Cabin microphone mixture $y(t)$
Optional playback reference $r(t)$
Optional echo cancellation + spatial front-end
Mixture encoder (multi-scale, Conv1D)
Conv1D (L1)
Conv1D (L2)
Conv1D (L3)
F_1
F_2
F_3
Speaker-conditioned TCN extractor (× 4 repeats)
M_1
M_2
M_3
Decoder (Conv1D)
Decoder (Conv1D)
Decoder (Conv1D)
$s_1(t)$ (to presence gate)
$s_2(t)$ (auxiliary)
$s_3(t)$ (auxiliary)
Target enrollment $x(t)$
Speaker encoder (multi-scale, shared weights)
Conv1D (L1)
Conv1D (L2)
Conv1D (L3)
E_1
E_2
E_3
Speaker embedding e
Training-only: speaker classification loss
Training-only: multi-scale SI-SDR loss
Target-presence gate
$s_1(t)$ (from SpEx+)
Scalar waveform CNN (default) | framewise log-mel PVAD + top-frame aggregation (optional config)
Presence probability p
p > threshold?
Optional enrollment / extracted-speaker similarity gate (when configured)
Target speech accepted?
No
Reject / Drop (non-target or no speech)
Yes
Wake-word detection
$s_1(t)$ (accepted target speech)
Downstream peak conditioning: attenuate only if peak > 0.95
1 s extracted-audio window
Standard KWS CNN (default); target-speaker KWS replaces it when configured
Optional secondary KWS (when configured)
Primary score $s1$
Secondary score $s2$
Score fusion (mean / min / max / geomean)
Temporal threshold + hit window + refractory
Wake detected?
No
Continue streaming
Yes
Command capture
From wake detection (when wake detected)
Wait for next accepted target speech (do not classify wake segment)
Reapply presence gate (as above)
Fixed 2 s capture with brief-pause timeout
Audio-to-intent
2 s target speech (from command capture)
Configured intent model: CRNN (default); wav2vec 2.0 when configured
Intent label
Label map
Vehicle action (e.g., climate, navigation, media)
Optional spoken reply

Using the interactive diagram

Hover over a linked block to reveal the PDF link cursor; click it to jump to its explanation. Use the return link at the end of each explanation to go back to the diagram.

Speaker extractor

The SpEx+ stage encodes the cabin mixture and the target enrollment at three temporal scales. The enrollment representation conditions a repeated temporal-convolution extractor, which estimates one mask per scale. Applying those masks to the mixture features and decoding them produces s_1, s_2, and s_3. Runtime inference forwards s_1; the other outputs support the multi-scale training objective.

Target-presence gate

The configured presence head estimates whether target speech is present in the extracted waveform. The scalar waveform CNN is the default; a framewise log-mel PVAD head with top-frame aggregation is available as a configured alternative. An optional speaker-similarity gate can additionally compare the enrolled and extracted speaker representations. A rejected segment is dropped before wake-word processing.

Primary wake-word detector

The standard KWS CNN is the default detector. When the enrollment-conditioned target-speaker KWS model is configured, it replaces the standard primary detector. The primary score is evaluated over the extracted audio window.

Secondary KWS and temporal decision

An optional secondary KWS model can contribute a second score. The pipeline combines primary and secondary scores using the configured fusion rule, then applies the wake threshold, temporal hit window, and refractory period. Without a wake decision, the stream continues.

Post-wake command capture

A wake event arms a separate command-capture state. The wake-word segment itself is not sent to intent classification. The system waits for the next accepted target-speech segment, rechecks presence, allows a brief pause, and resets capture after the configured timeout.

Audio-to-intent classifier

The next accepted command segment is prepared as a fixed two-second input. The CRNN is the default classifier; a wav2vec 2.0 encoder and classifier head can be selected by configuration. The predicted intent class is mapped to a vehicle action and may include a spoken reply.

Training-only paths

The dotted connections in the diagram identify training supervision. Multi-scale SI-SDR loss supervises the extracted outputs, while speaker-classification loss supervises the speaker representation. These losses are not part of runtime inference.

Return to the diagram